Project assignment

Deadline (beginning of February)

Project

- **Design and implement an adaptive version of the CSMA/CA protocol using a Reinforcement Learning** (RL) **technique**
  - multi-armed bandit
  - Q-learning
- Each node in the network should learn from its own experience — by observing collisions and successful transmissions — and adapt its contention behavior to improve overall performance

Steps (1/2)

1. Implement basic CSMA/CA (static baseline)
2. Implement a RL-based version (each node is an independent agent)
   - multi-armed bandit (e.g., ε-greedy or UCB)
   - a Q-learning algorithm
3. Run simulations under different network conditions
   - Vary the number of nodes and the packet generation probability
   - Compare the adaptive protocol with the static CSMA/CA baseline
   - Vary the exploration rate (ε) and analyze impact on performance

Steps (2/2)

4. Metrics
   - Packet delivery ratio: how many packets arrive successfully compared to those generated
   - Throughput: amount of useful data transmitted per time unit
5. Analyze and discuss:
   - Does the learning algorithm reduce collisions?
   - Does it improve throughput ?
   - Other observation?
6. Write a report including
   - Description of the model and assumptions
   - Explanation of the learning algorithm and parameters
   - Experimental setup and results
   - Discussion and conclusions
7. Show a demo